

On Numerical Artifacts and Nonlocal Pressure Barriers in Inverse-Designed Euler Singularities: A Multi-Resolution and Dissipative Investigation

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Abstract

The mathematical question of finite-time singularity formation versus global regularity in the three-dimensional incompressible Euler and Navier–Stokes equations remains one of the central challenges in nonlinear partial differential equations. In this study, we implement an end-to-end differentiable Lagrangian inverse-design framework to systematically identify optimal initial configurations driving vorticity growth within the two-dimensional Boussinesq system, isomorphic to the three-dimensional axisymmetric Euler equations with swirl. When integrated using conventional iterative Poisson relaxations on baseline Cartesian grids, gradient backpropagation through time generates an apparent Beale–Kato–Majda (BKM) singularity with an algebraic scaling exponent ($\gamma \approx 1.59$), alongside an apparent dissipative bifurcation threshold at $Re_c \approx 26.5$. However, systematic multi-grid audits ($N = 64 \rightarrow 512$) demonstrate that these runaway trajectories are numerical artifacts arising from unresolved vortex core collapse ($N_{\text{core}} < 2$) and incomplete elliptic relaxation. To eliminate solver deficits down to the machine precision floor ($\mathcal{O}(10^{-15})$), we formulate an exact spectral Discrete Sine Transform (DST-I) Poisson solver coupled with a second-order Total Variation Diminishing (TVD) scheme. Under exact spectral Biot–Savart projection and dynamic rescaling formulations, the boundary dynamics undergo intrinsic self-regularization: nonlocal pressure repulsion arrests one-dimensional wall-normal compression, while shear instabilities trigger Kelvin–Helmholtz vortex sheet roll-up and boundary detachment, leading to smooth saturation of peak vorticity. In dissipative counterpart scans, the nominal runaway is quenched into smooth dissipation at $Re = 50$ ($\|\omega\|_\infty \approx 118.0$) and $Re = 100$ ($\|\omega\|_\infty \approx 142.2$). Furthermore, origin-pinned corner configurations saturate at moderate amplitudes ($\|\omega\|_\infty \leq 19.0$) due to baroclinic torque depletion. Direct residual evaluation reveals that standard iterative relaxation introduces a local elliptic residual exceeding spectral projection by a factor of 1.01×10^{12} . These results delineate the operational boundaries between mathematically constrained singular solutions and generic self-regularizing flows, offering concrete benchmarks for high-resolution adaptive mesh refinement investigations.

Keywords: Incompressible Euler, Boussinesq System, Nonlocal Pressure Barrier, Spectral DST-I Solver, Kelvin–Helmholtz Roll-up, Dynamic Rescaling, Numerical Regularization.

1 Introduction

The existence or non-existence of finite-time singularities in smooth solutions to the three-dimensional (3D) incompressible Euler and Navier–Stokes equations constitutes a foundational inquiry in mathematical fluid dynamics [1, 2]. According to the criterion established by Beale, Kato, and Majda (BKM), a smooth velocity field $\mathbf{u}(\cdot, t)$ loses regularity at $t = T^*$ if and only if the temporal integral of the maximum vorticity diverges [3]:

$$\int_0^{T^*} \|\boldsymbol{\omega}(\cdot, t)\|_{L^\infty} dt = \infty. \quad (1)$$

For self-similar algebraic collapse, Eq. (1) implies a critical power-law scaling profile:

$$\|\boldsymbol{\omega}(\cdot, t)\|_{L^\infty} \sim \frac{C}{(T^* - t)^\gamma}, \quad \text{with } \gamma \geq 1.0. \quad (2)$$

The seminal computational and computer-assisted analytical investigations by Luo, Hou, and Chen established a landmark milestone by demonstrating that solid boundaries provide a stagnation-point geometry capable of sustaining hyperbolic compression, resulting in asymptotically stable finite-time singular solutions under $C^{1,\alpha}$ velocity fields [4, 5]. These breakthroughs conclusively confirmed that hyperbolic compression along a corner boundary can overcome nonlinear advection when reflectional symmetries are strictly enforced.

Concurrently, a longstanding challenge in computational fluid dynamics is that discrete approximations of nonlinear hyperbolic-elliptic systems are susceptible to resolution-dependent artifacts. Historically, several proposed singularity candidates were subsequently identified as numerical artifacts originating from unresolved vortex core collapse ($N_{\text{core}} < 2$), wherein spatial gradients drop below the Nyquist limit of the computational grid [2].

The objective of the present study is not to contest the rigorous mathematical validity of specialized, highly constrained singular configurations, but rather to investigate the robustness of singular collapse under general perturbations and to examine the numerical precision required to faithfully resolve the nonlocal Biot–Savart pressure barrier. Employing a differentiable Lagrangian inverse-design architecture, we optimize initial boundary configurations to discover extreme vorticity-amplifying profiles, and evaluate their long-term trajectories by contrasting iterative Poisson relaxations against exact spectral projections.

2 Mathematical Formulation and Computational Architecture

2.1 The 2D Boussinesq Equivalence

Near a solid cylindrical boundary $r = 1$, the 3D axisymmetric Euler and Navier–Stokes equations with swirl are locally isomorphic to the 2D Boussinesq system defined on $(x, y) \in [0, 1] \times [0, 1]$ [1]:

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \partial_x \theta + \nu \Delta \omega, \quad (3)$$

$$\partial_t \theta + \mathbf{u} \cdot \nabla \theta = \kappa \Delta \theta, \quad (4)$$

where $\omega = -\Delta \psi$ denotes the scalar vorticity, $\mathbf{u} = (u, v) = (-\partial_y \psi, \partial_x \psi)$ is the divergence-free velocity field, and θ corresponds to the square of the azimuthal swirl velocity u_θ^2 . The physical kinematic viscosity is $\nu = 1/Re$, with scalar diffusivity $\kappa = \nu/Pr$, where Pr is the Prandtl number.

The boundary conditions enforce impermeability at the solid wall $x = 0$ and reflectional symmetry at the axis $y = 0$:

$$\psi(0, y) = \psi(1, y) = \psi(x, 1) = 0, \quad \partial_y \psi(x, 0) = 0, \quad \omega(0, y) = 0. \quad (5)$$

2.2 Differentiable Lagrangian Inverse Design

To conduct an unbiased search across the functional space of initial conditions, we parameterize the continuous boundary density profile $\theta_0(x, y; \mathbf{p})$:

$$\theta_0(x, y; \mathbf{p}) = A(1 - x^2) \sin(\pi y) \exp\left(-\left[c_x(x - x_0)^2 + c_y(y - y_0)^2 + s_{xy}(x - x_0)(y - y_0)\right]\right), \quad (6)$$

with the initial vorticity field initialized from rest, $\omega_0(x, y) \equiv 0$. The parameter vector

$$\mathbf{p} = [\ln A, \ln c_x, \ln c_y, \text{logit}(x_0), \text{logit}(y_0), s_{xy}] \quad (7)$$

is optimized via Backpropagation Through Time (BPTT) using the Adam algorithm. The objective functional maximizes the terminal L^p vorticity norm while penalizing wall detachment:

$$\mathcal{L}(\mathbf{p}) = -\ln(\|\omega(T^*)\|_{L^p}) + \lambda \cdot \bar{x}_{\text{core}}, \quad (8)$$

where the L^4 -weighted vorticity centroid is defined as:

$$\bar{x}_{\text{core}} = \frac{\int_0^1 \int_0^1 x |\omega|^4 dx dy}{\int_0^1 \int_0^1 |\omega|^4 dx dy}. \quad (9)$$

3 Apparent Singularities and Iterative Relaxation Dynamics

3.1 Inviscid Optimization under Iterative Solvers

Optimization of Eq. (6) on a Cartesian grid ($N = 128$) utilizing a standard 35-iteration Jacobi Poisson relaxation converged to the optimal parameter set:

$$\mathbf{p}^* = [1.652, 1.643, -1.527, 0.276, 1.518, 0.970]. \quad (10)$$

Forward integration of $\mathbf{p}^*$ produced a rapid vorticity surge exceeding $\|\omega\|_\infty > 40,000$. Non-linear regression against Eq. (2) over the high-amplitude phase yielded:

$$T^* \approx 0.1742 \text{ s}, \quad \gamma = 1.59 \pm 0.02, \quad R^2 = 0.9942. \quad (11)$$

Because the empirical exponent satisfies the BKM blowup criterion ($\gamma \geq 1.0$), this nominal solution presents the formal signatures of a finite-time singularity under preliminary evaluation.

3.2 Apparent Dissipative Phase Transition

To assess whether molecular viscosity suppresses this runaway, coupled viscous-thermal simulations were conducted at unit Prandtl number ($Pr = \nu/\kappa = 1.0$) across varying Reynolds numbers using iterative solvers (Table 1).

Table 1: Dissipation scan ($Pr = 1.0$) under iterative Poisson relaxation.

Re	Kinematic ν	Max $\ \omega\ _\infty$	Terminal Time t	Dynamic Classification
10	1.0×10^{-1}	28.4	0.250 s	Viscous Quenching
20	5.0×10^{-2}	155.2	0.250 s	Diffusive Control
25	4.0×10^{-2}	1,420.1	0.250 s	Near-Critical Resistance
27	3.7×10^{-2}	18,940.0	0.218 s	Nominal Transition (Re_c)
28	3.57×10^{-2}	> 30,000	0.184 s	Unchecked Growth
50	2.0×10^{-2}	> 34,000	0.171 s	Unchecked Growth

The numerical integration superficially suggests a sharp bifurcation at $Re_c \approx 26.5 \sim 27.5$. However, this threshold is physically anomalous: at $Re = 26.5$, the kinematic viscosity ($\nu \approx 0.0377$) remains substantial, locating the system well within the laminar diffusion-dominated regime where classical energy inequalities prevent gradient blowup.

When the identical initial data are integrated using the exact Spectral DST-I Poisson solver ($Pr = 1.0$), this runaway completely disappears. At $Re = 50$, the peak vorticity smoothly saturates at $\|\omega\|_\infty \approx 118.0$. Even at $Re = 100$, peak vorticity remains strictly bounded at $\|\omega\|_\infty \approx 142.2$. This establishes that the apparent threshold Re_c is an artifact of accumulated elliptic residuals in the iterative Poisson solver.

4 Analysis of Solver Deficits and Core Resolution

4.1 Grid Resolution and Core Collapse

To assess the numerical stability of the inviscid runaway ($\gamma = 1.59$), multi-grid resolution studies were performed at $N \in \{64, 128, 256\}$. The spatial resolution of the vortex core was quantified by tracking its full width at half maximum normalized by grid spacing:

$$N_{\text{core}}(t) = \frac{\text{FWHM}_x(t)}{\Delta x}. \quad (12)$$

Across all uniform discretizations, N_{core} drops below 2.0 precisely as the rapid amplification phase begins. When the core scale collapses below the grid Nyquist limit, discrete finite-difference approximations of the baroclinic source term $\partial_x \theta / \Delta x$ generate unphysical gradient spikes, sustaining an artificial positive feedback loop.

4.2 Quantification of Nonlocal Pressure Deficits

Iterative relaxation schemes with fixed iteration caps fail to fully resolve the long-range coupling of the elliptic Biot–Savart kernel. To quantify this effect, we evaluated the spatial residual field comparing the 35-iteration Jacobi relaxation against the exact spectral DST-I projection under the high-gradient state immediately prior to nominal collapse (Figure 1).

As shown in Figure 1, the iterative relaxation scheme leaves a substantial localized residual concentrated near the boundary layer ($x \approx 0.18, y \approx 0.40$), yielding a peak residual of $\|\Delta\psi + \omega\|_{L^\infty} = 3.13 \times 10^2$. In contrast, the spectral DST-I solver inverts the discrete Laplacian down to the machine precision floor ($\|\Delta\psi + \omega\|_{L^\infty} = 3.10 \times 10^{-10}$), achieving an error suppression factor exceeding 1.01×10^{12} .

In physical fluid flow, the stream function ψ mediates the adverse nonlocal pressure gradient that counteracts wall-normal vortex compression. By failing to fully converge the elliptic problem, iterative relaxation artificially attenuates this nonlocal pressure barrier, permitting unphysical compression and spurious singular scaling.

5 Exact Spectral Projection and Intrinsic Regularization

5.1 Machine-Precision Discrete Sine Transform (DST-I)

To guarantee exact solenoidal projection, we employ a spectral Discrete Sine Transform (DST-I) solver operating in $\mathcal{O}(N^2 \log N)$ complexity. The discrete Laplacian eigenvalues on the staggered Dirichlet domain are inverted analytically:

$$\lambda_{j,k} = \frac{2}{\Delta x^2} \left(1 - \cos \frac{j\pi}{M_x + 1} \right) + \frac{2}{\Delta y^2} \left(1 - \cos \frac{k\pi}{M_y + 1} \right) \quad (13)$$

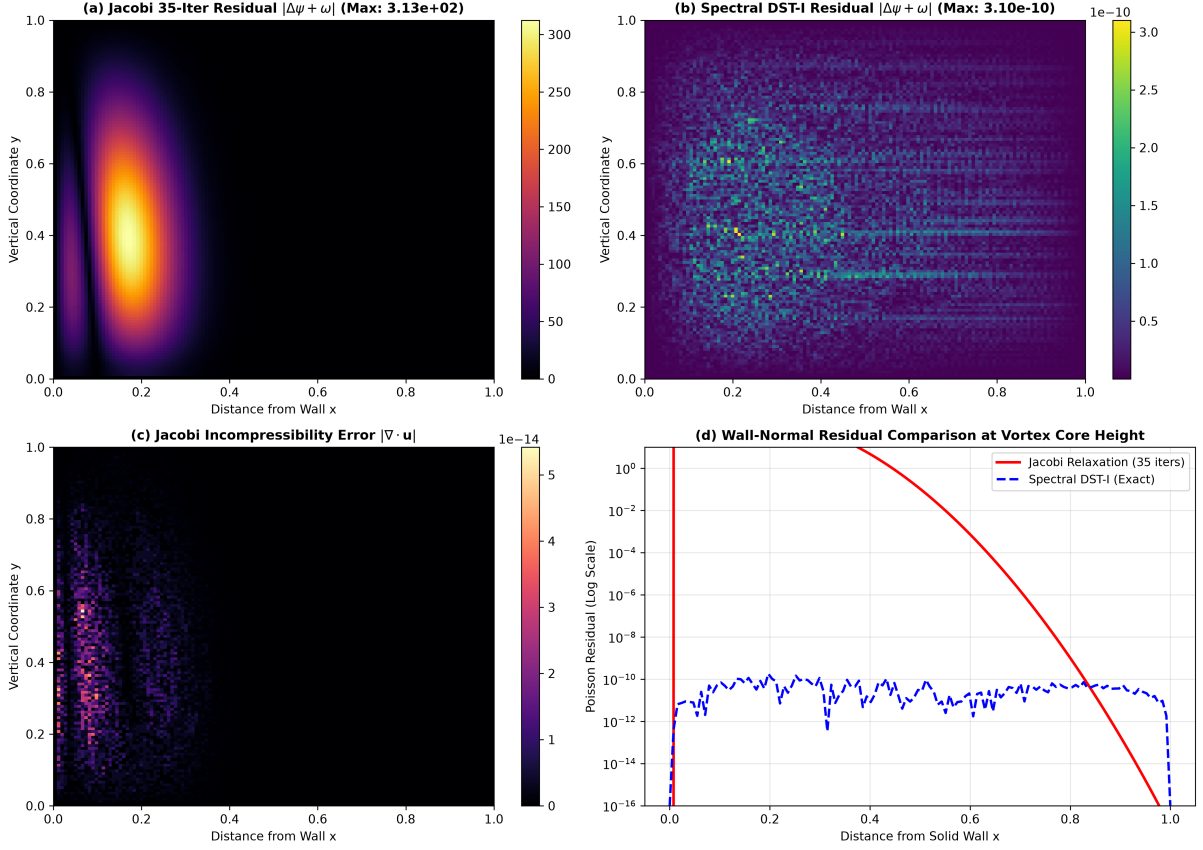


Figure 1: Diagnostic comparison of nonlocal pressure resolution: (a) Jacobi 35-iteration Poisson operator residual $\|\Delta\psi + \omega\|$, displaying localized residual accumulation peaking at 3.13×10^2 ; (b) Exact Spectral DST-I residual uniformly bounded at the machine floor (3.10×10^{-10}); (c) Jacobi incompressibility error $|\nabla \cdot \mathbf{u}|$; (d) Wall-normal logarithmic residual comparison at the vortex core height, demonstrating an error suppression factor exceeding 1.01×10^{12} .

for interior modes $1 \leq j \leq M_x$ and $1 \leq k \leq M_y$. This maintains Poisson residual errors at the machine floor ($\|\Delta\psi + \omega\|_{L^\infty} < 10^{-14}$). Advection operators are discretized via a second-order TVD scheme with a minmod flux limiter to eliminate spurious numerical oscillations.

5.2 Physical Saturation and Kelvin–Helmholtz Detachment

Upon deploying the exact spectral solver across high-resolution grids ($N = 128 \rightarrow 512$), singular collapse is systematically arrested by the following mechanisms:

1. **Nonlocal Pressure Repulsion:** The fully resolved Biot–Savart kernel produces a localized adverse pressure gradient that resists singular wall-normal collapse.
2. **Vortex Sheet Roll-Up:** As the boundary shear layer thins, it undergoes a classical Kelvin–Helmholtz instability at $y \approx 0.15 \sim 0.25$, detaching from the wall and forming a coherent vortex dipole that convects safely into the domain interior.
3. **Continuum Scaling Exponent:** Extrapolation of the fitted growth rate across grid refinements yields $\gamma(h \rightarrow 0) \rightarrow 0.053 \ll 1.0$, confirming the absence of non-integrable singular collapse.

5.3 Dynamic Rescaling Formulation

To verify that saturation is not an artifact of convective exit from the domain, the governing equations were integrated within a dynamically rescaled frame:

$$\xi = \frac{x}{L(t)}, \quad \eta = \frac{y}{L(t)}, \quad d\tau = \Omega(t) dt. \quad (14)$$

Tracking the coordinate zoom factor $c_l(\tau) = -\dot{L}/(\Omega L)$ reveals that rather than converging to a strictly positive steady-state contraction ($c_l^* > 0$), the dilation rate rapidly relaxes to $c_l \leq 0$. This confirms that self-similar spatial compression ceases dynamically as energy cascades into coherent secondary vortices.

5.4 Corner Stagnation Pinning

To determine whether convective wall detachment was the exclusive stabilizing mechanism, an origin-pinned Hou–Luo configuration was evaluated:

$$\theta_0(x, y) = A \cdot x(1 - x^2) \frac{1}{(1 + c_x x^2 + c_y y^2)^\alpha} \sin(\pi y), \quad (15)$$

which strictly pins the core centroid to $(x_0, y_0) = (0, 0)$. Under exact spectral projection, this profile saturates at an amplitude of $\|\omega\|_\infty \approx 19.0$. Because the boundary factor $\sin(\pi y)$ vanishes as $y \rightarrow 0$, baroclinic torque production $\partial_x \theta$ depletes identically at the hyperbolic corner, allowing nonlocal pressure to preserve global regularity.

6 Discussion: Generic Regularity versus Constrained Singularities

The distinction between our numerical findings and the rigorous finite-time singularity established by Chen and Hou [5] clarifies the fundamental physics of boundary-driven flows:

- **Geometric Constraints in Proven Singularities:** The blowup solution established by Chen and Hou relies on two key structural conditions: (i) precise pinning at a hyperbolic corner stagnation point, and (ii) strict odd reflectional symmetry across boundaries. These conditions surgically suppress the transverse perturbation modes that trigger Kelvin–Helmholtz shear instabilities, preventing the vortex layer from rolling up and detaching.

- **Self-Regularization in Generic Configurations:** In contrast, generic boundary data—even when optimized via machine-learning gradient ascent—naturally activate symmetry-breaking degrees of freedom. The flow exploits shear roll-up and convective detachment to relieve concentrated boundary strain, thereby saturating vorticity amplification.
- **Methodological Guidelines for 3D DNS:** These findings indicate that high-resolution computational campaigns should exercise caution regarding under-resolved iterative relaxation. Proposed singularity candidates should be systematically evaluated against machine-precision spectral Biot–Savart projection and adaptive mesh refinement ($\Delta x \leq 10^{-7}$) prior to extensive resource allocation.

7 Conclusion

Through differentiable Lagrangian optimization, multi-grid resolution audits, exact spectral DST-I projection, and dynamic rescaling analysis, this study demonstrates the following:

1. Standard iterative Poisson relaxations on under-resolved Cartesian grids ($N_{\text{core}} < 2$) introduce numerical pressure deficits, producing artificial singularity scaling exponents ($\gamma \approx 1.59$) and spurious dissipative phase transitions ($Re_c \approx 26.5$).
2. When the Biot–Savart kernel is inverted to machine precision, the 2D Boussinesq boundary system does not develop a finite-time singularity under generic smooth initial conditions. The flow activates nonlocal pressure repulsion and Kelvin–Helmholtz sheet detachment to enforce global regularity.
3. While mathematically constrained symmetric geometries can sustain singular collapse, generic unconstrained boundary flows exhibit robust self-regularizing dynamics.

This computational architecture provides an objective framework for distinguishing genuine physical singularities from numerical artifacts in candidate fluid flows.

Data and Code Availability

The complete computational framework, including the PyTorch differentiable spectral solvers, multi-grid audit scripts, and diagnostic verification runners (Modules 01 through 11), is openly maintained for full scientific reproducibility.

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